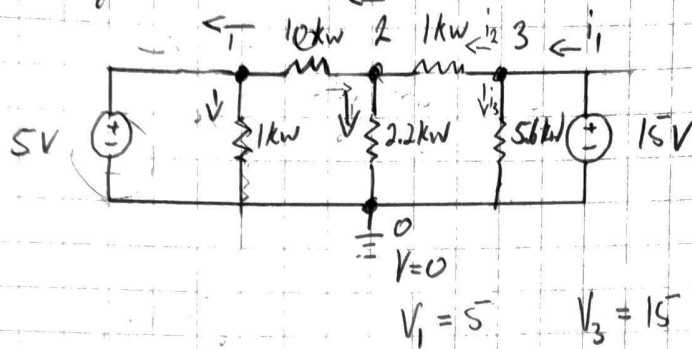


Fig 3.1 Resistive Circuit



$$V_{5.6} = 15V - 0V = 15V$$

$$V_{1k} = 5V - 0V = 5V$$

$$V_{1R} = 15V - V_2 = 5.03V$$

$$V_2 = \frac{V_2}{2.2} + \frac{V_2 - 5}{10} = \frac{15 - V_2}{1} \rightarrow 10V_2 + 2.2(V_2 - 5) = 22(15 - V_2)$$

$$V_{2.2} = \frac{9.97V}{2200\Omega} = 4.53mA$$

$$V_3 = 15$$

$$i_3 = \frac{15V}{5.6} = 2.68mA$$

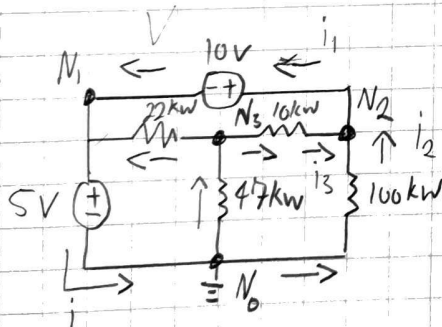
$$i_2 = \frac{5.03V}{1000} = 5.03mA$$

$$10V_2 + 2.2V_2 - 11 = 330 - 22V_2$$

$$34.2V_2 = 341$$

$$V_2 = 9.97076V \approx 9.97V$$

$$V_{10} = V_2 - V_1 = 9.97V - 5 = 4.97V$$



$$V_1 = 5V$$

$$V_2 = V_1 + 10 = 15V$$

$$i_1 = i_2 + i_3$$

$$N_3 \Rightarrow \frac{V_3 - V_1}{22} + \frac{V_3 - V_2}{10} = \frac{-V_3}{47}$$

$$10V_3 - 10V_1 + 22V_3 - 22V_2 = (22)(10) \frac{-V_3}{47}$$

$$47(10V_3 - 50 + 22V_3 - 330) = (22)(10)(-V_3)$$

$$1504V_3 - 17860 = -220V_3$$

$$1724V_3 = 17860$$

$$V_3 = 10.3596V$$

$$V_{22} = V_3 - V_1 = 5.36V$$

$$V_{10} = V_2 - V_3 = 4.64V$$

$$V_{47} = V_3 = 10.36V$$

$$V_{100} = V_2 = 15V$$